Advances in understanding and controlling rice blast disease: Mechanisms and management strategies

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Highlights

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Magnaporthe oryzae causes devastating rice blast disease, affecting all growth stages.

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Advances in chemical and biological control, resistance breeding, and molecular strategies.

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Emerging approaches including gene pyramiding, HIGS/SIGS, and CRISPR/Cas-based genome editing.

Abstract

Rice blast, caused by the filamentous fungus *Magnaporthe oryzae*, is one of the most destructive diseases affecting rice cultivation worldwide, posing a serious threat to food security and economic stability. The disease can infect rice at all developmental stages, from seedling emergence to grain filling, leading to yield losses and even total crop failure. This review summarizes recent advances in rice blast research, with a focus on disease symptoms, causal factors, and control strategies. We highlight progress in chemical and biological control, resistance breeding, and molecular approaches, with particular emphasis on genome editing, RNA interference technologies, effector biology,

and omics-guided functional genomics. Emerging tools such as CRISPR/Cas-based editing, host- and spray-induced gene silencing, genomics, and AI-assisted resistance design are reshaping blast resistance discovery and deployment. By integrating genetic resistance with climate-adaptive cultivation, a prevention-oriented and sustainable management framework is proposed. This comprehensive overview provides insights into current challenges and future directions for rice blast control.

Graphical abstract

Keywords

Magnaporthe oryzae

Rice blast

Biocontrol

Molecular approaches

1. Introduction

Rice (*Oryza sativa* L.) serves as the primary food source for more than half of the world's population, sustaining over 3.5 billion people daily [1]. As a key source of global food security, stable rice production is critical for social and economic stability, particularly in Asia, where over 90 % of the world's rice is produced and consumed [2,3]. However, rice blast, caused by the hemibiotrophic ascomycete fungus *Magnaporthe oryzae*, remains the most devastating disease of rice worldwide [4]. Often referred to as “rice fever” or “kneeling blast” due to its capacity to decimate fields rapidly, the disease has been reported in more than 85 rice-producing countries across Asia, Africa, Latin America, and Europe, and continues to expand its geographic and ecological range [5,6]. South-East Asia is widely recognized as the center of origin, genetic diversity, and global dissemination of *M. oryzae*, underscoring the region's pivotal role in the emergence, evolution, and spread of rice blast epidemics [7].

Globally, rice blast disease causes approximately 30 % of the annual yield loss. In severe outbreaks, it can reduce the yield by up to 50 % or even lead to complete crop failure. In high-production areas such as China, India, and Southeast Asia, localized outbreaks can result in losses of millions of tons of grain per year, destabilizing food markets. To prevent the outbreak of blast disease, a ‘‘one-vote veto” for breeding rice with blast resistance has been implemented in rice cultivars certification in China since 2008 [8]. This means that rice cultivars showing poor blast disease resistance will not get official certificate to allow for selling as seeds to farmers for growing.

M. oryzae exhibits remarkable ecological adaptability, allowing it to infect rice plants at nearly all growth stages, including leaf, node, neck, panicle, and grain [9]. The success of this pathogen as a pathogen is attributed to its ability to rapidly evolve new physiological types, as well as its sensitivity to environmental factors such as humidity, temperature fluctuations, nitrogen fertilizer application, and crop rotation [10,11]. This adaptability poses significant obstacles to long-term resistance breeding and rice blast disease management. In recent years, changes in climate patterns, including rising temperatures and alterations in rainfall patterns, have further complicated the complexity of rice disease outbreaks.

Conventional control methods, such as the use of fungicides and the cultivation of disease-resistant varieties, have achieved certain levels of success. While chemical fungicides, such as tricyclazole and azoxystrobin, can suppress disease outbreaks, their widespread use has raised concerns regarding the environment and health, leading to the development of resistance in pathogens [12,13]. Similarly, the deployment of resistance genes such as *Pi1*, *Pi2*, *Piz-t*, and *Pigm* in breeding programs has enhanced the disease control effect; however, due to the rapid evolution of harmful rice blast strains, this resistance is often temporary [14]. To address these limitations, recent studies have focused on innovative and sustainable solutions, including gene editing technologies like CRISPR/Cas9, host-induced gene silencing (HIGS), RNA interference (RNAi), microbial biocontrol agents, and precision agriculture techniques [[15], [16], [17]]. These approaches aim to provide more durable, environmentally friendly alternatives to traditional control measures.

Here, we briefly summarize the latest research findings in the fields of rice blast disease epidemiology, pathogenesis, and control strategies. We emphasize the progress made and also point out the remaining challenges.

2. Symptoms of rice blast

Rice blast disease manifests in six different forms during the various growth and development stages of rice plants. Each form affects specific parts and growth stages of the plants, which reflects the strong adaptability of the pathogen causing wheat blast disease (Fig. 1).

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Fig. 1. Symptoms of rice blast disease across rice plant parts and growth stages. The diagram illustrates the major symptomatic manifestations of rice blast (Seedling blast, Leaf blast, Nodal blast, Leaf collar blast, Neck blast, and Grain blast), one of the most devastating fungal diseases of rice worldwide, caused by *M. oryzae*. The central illustration shows a mature rice plant with red boxes highlighting infection sites. Surrounding circular insets provide close-up photographs of characteristic lesions. These symptoms vary by plant stage, environmental conditions (high humidity, moderate temperature), and pathogen race, emphasizing the need for integrated management across the crop cycle. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.1. Seedling blast

Seedling blast primarily occurs during the three-leaf stage and is also associated with the presence of *M. oryzae* in seeds. Early infections can lead to yellowing or browning of seedlings, with the base turning grayish-black, resulting in plant death [18]. During the three-leaf stage, characteristic spindle-shaped or rhomboid lesions appear, sometimes accompanied by irregular small spots. In high-humidity environments, a gray mold layer forms at the base of the seedling, and the upper parts turn light reddish-brown, causing the seedling to die. Notably, recent studies have emphasized the role of seed treatments with biocontrol agents, such as *Bacillus subtilis*, in reducing seedling blast incidence by up to 50 % in controlled settings [19]. This indicates that detecting and removing contaminated seeds can help reduce the early infection of rice seedlings.

2.2. Leaf blast

Leaf blast is prevalent during the tillering stage, but it can occur throughout the entire growth cycle of rice. It symptoms vary depending on the variety's resistance, environmental conditions, and the pathogen's virulence. Lesions can be classified as chronic, brown spot, white spot, and acute types. Among them, the acute type has the strongest destructive power due to rapid spread [10]. High humidity and moderate temperatures ranging from 20 to 30 °C promote spore germination and the development of lesions. A recent study has emphasized the importance of using hyperspectral imaging for early detection of leaf blast, enabling timely interventions [20]. Li et al. [21] pointed out that the duration of leaf wetness is a key determinant of successful infection, and the minimum duration of required wetness is also related to temperature. These findings provide a basis for current irrigation management strategies aimed at reducing long-term leaf surface humidity to lower the risk of early-stage leaf spot disease infection.

2.3. Nodal blast

After rice heading, if there is too much or too little water and nutrients, nodal blast is likely to occur. The affected parts are usually 1–2 nodes below the panicle. When the disease first occurs, the main symptom is the appearance of small brown lesions on the stem nodes, followed by the gradual expansion of the lesions until the stem breaks and the rice panicle dries and dies [9]. If the lesions are concentrated on one side of the node, it may lead to rice deformation.

2.4. Leaf collar blast

This kind of blast is generally rare, often occurring at the junction of the leaf sheath and leaf blade. In the early stage of the disease, irregular grayish-green spots appear on the leaves, and in the late stage, the diseased leaves will break off at the leaf collar. When leaf collar blast is severe, it can spread to the flag leaf collar, easily leading to neck blast.

2.5. Neck blast

Neck blast is one of the most economically devastating forms of rice blast, affecting the panicle neck and rachis during the heading stage. Initial lesions are spindle-shaped and light brown, but gradually darken to black as they expand along the panicle axis. Infections occurring at early heading often result in white panicles or shriveled grains, whereas infections at later stages mainly reduce seed quality and grain weight. Yield losses due to neck blast can be severe, reaching up to 70 % in highly susceptible cultivars under epidemic conditions [22]. Recent advances in high-throughput monitoring, particularly the use of unmanned aerial vehicles (UAVs) and multispectral imaging, have greatly improved the precision of early disease detection, thereby facilitating timely and targeted fungicide applications [23].

2.6. Grain blast

Grain blast affects mature or semi-mature grains, predominantly attacking the hull or lemma. Initial infections manifest as dark brown or grayish-white spots, which may eventually coalesce and blacken the entire grain surface. When infection occurs at later stages, the lemma often turns brown, resulting in discolored or shriveled seeds that act as a major source of primary inoculum for the subsequent cropping season. Recent advances in seed health management have demonstrated that plasma-based seed disinfection and biocontrol agent coatings can significantly reduce grain blast transmission, lowering seedborne inoculum by up to 60 % under experimental conditions [24]. In addition, molecular diagnostic tools—particularly qPCR assays—have enhanced the sensitivity and reliability of seed health testing, thereby facilitating the production and use of pathogen-free planting material [25].

3. Causes of rice blast occurrence

The occurrence and severity of rice blast, caused by the fungus *M. oryzae*, result from the interaction of four primary factors: the host (rice plant), the pathogen, environmental conditions, and agronomic management practices. This classic disease tetrahedron framework highlights how imbalances in these elements create favorable conditions for disease outbreaks

Fig. 2. Factors causing rice blast disease in rice plants. It highlights the interactions among the host, pathogen, and cultivation management, and climate environment that influence disease development.

3.1. Host susceptibility and pathogen virulence

Rice blast epidemics are driven by a dynamic coevolutionary arms race between the rice host and the pathogen *M. oryzae*. Host susceptibility is primarily determined by the genetic makeup of rice varieties, many of which—particularly traditional or high-yielding types—lack broad-spectrum resistance. Even varieties carrying major resistance genes often experience rapid breakdown when deployed over large areas, typically within 3–5 years, as new virulent pathogen races emerge through selection pressure [4,21]. On the pathogen side, *M. oryzae* maintains high-virulence strains capable of overcoming specific resistance genes, supported by extensive genetic diversity revealed through population genomic studies and remarkably rapid evolutionary adaptation to host defenses [26]. This interplay underscores the challenge of durable resistance; recent advances in gene pyramiding—stacking multiple resistance genes—and genomic selection strategies aim to broaden and prolong host resistance by anticipating pathogen evolution, enabling more proactive and dynamic breeding approaches.

3.2. Cultivation management

Cultivation practices play a critical role in shaping the field microenvironment and host physiological status, thereby strongly influencing rice blast development. Cold, waterlogged soils with poor aeration favor the survival and spread of *M. oryzae*, whereas optimized planting patterns such as wide–narrow row spacing and semi-dry cultivation improve canopy ventilation, light penetration, and photosynthetic efficiency, collectively reducing disease incidence [4]. Recent research advocates for conservation tillage combined with cover crops to suppress soil-borne inoculum [27], while precision tillage, guided by soil health sensors and real-time monitoring, has been reported to reduce blast incidence by optimizing soil aeration, nutrient cycling, and beneficial microbial activity [28]. Planting density is another key determinant, as excessive density leads to canopy

closure, poor air circulation, sustained high humidity (>90 %), and reduced evapotranspiration, creating favorable conditions for spore germination and infection [29]. Overcrowding also weakens host resistance by reducing photosynthetic efficiency, nitrogen-use efficiency, and overall plant vigor, thereby increasing susceptibility, particularly during the tillering stage [30]. Fertilization and irrigation practices further modulate blast risk by altering host resistance and microclimatic conditions. Excessive or poorly timed nitrogen application lowers the C/N ratio, reduces silicification, and increases free amino acid accumulation, all of which enhance host susceptibility to *M. oryzae*. In contrast, balanced applications of phosphorus and potassium strengthen cell walls and enhance resistance against *M. oryzae* [31], while silicon fertilization enhances the silicification of rice stems and leaf epidermis, forming a physical barrier that impedes pathogen invasion and reducing blast severity by 30–50 %, with concomitant reductions in pesticide use [32,33].

Improper irrigation practices (e.g., prolonged flooding and excessive field humidity) extend leaf wetness duration, thereby promoting spore germination and pathogen infection. Appropriate irrigation management should avoid waterlogging, maintain adequate ventilation and light penetration, and reduce humidity levels in the field [34]. Furthermore, sensor-based irrigation scheduling has enhanced water-use efficiency and reduced blast incidence by enabling precise control of soil moisture dynamics [35].

3.3. Climatic conditions

Climatic factors, particularly temperature, high humidity, and prolonged cloudy weather, create ideal conditions for *M. oryzae* spore germination and dispersal [36]. Low light intensity weakens plant vigor, while wind facilitates spore spread, though dry, sunny conditions inhibit pathogen activity [37]. Recent studies have increasingly focused on the impacts of climate change on rice blast epidemiology. Climate projection models predict that rising global temperatures, altered precipitation patterns, and increased frequency of extreme weather events will significantly elevate rice blast risk in many major rice-growing regions [38,39]. For example, modeling analyses suggest that warming trends combined with erratic rainfall may extend the blast-conducive season and expand the geographic range of severe epidemics. Therefore, studies integrate weather forecasting with disease prediction models, enabling preemptive control measures [40].

4. Rice blast control

Effective rice blast management integrates chemical control and biocontrol, genetic and molecular strategies, with recent innovations enhancing their efficacy and sustainability (Fig. 3).

Fig. 3. Integrated strategies for rice blast control. The diagram contrasts traditional and emerging strategies for controlling rice blast (*M. oryzae*). (A) It highlights the primary role of molecular breeding (CRISPR editing, HIGS/SIGS, omics) and resistant varieties (*R* genes pyramiding) in rice blast control (B and C) Possible pathways of chemical control (e.g., Tricyclazole, Propiconazole mixtures). and biocontrol agents (plant/microbe-derived biopesticides and phytoalexins) are shown. (D) The potential directions of climate-adaptive management. (E) Potential model for integrated management for rice blast controls in the future.

4.1. Chemical control and biocontrol

4.1.1. Chemical control

Chemical control remains one of the most widely used methods for integrated rice blast management due to its relative ease, low cost, and high efficacy. Historically, copper- and mercury-based preparations were widely applied from the 1950s, inhibiting fungal enzymes involved in glycolysis and energy production, ultimately killing the pathogen [41]. A recent study has shown that certain genes enhance resistance to *M. oryzae* by promoting the uptake of copper [42]. However, these fungicides often cause phytotoxicity—leaf scorching, reduced photosynthesis, and stunted growth—especially under high humidity or acidic soil conditions [43]. Following the limitations and bans on copper and mercury preparations, research shifted toward safer, more targeted chemical alternatives.

With the development of modern fungicides, chemical control strategies have evolved from traditional copper- and mercury-based formulations to more targeted single-component and compound fungicides, each with distinct mechanisms, efficacy, and limitations. Single-component fungicides, such as tricyclazole, Isoprothiolane, Tebuconazole, and Kasugamycin, are commonly used today. Tricyclazole, a thiazole fungicide, inhibits spore germination and appressorium formation and is rapidly absorbed by roots, stems, and leaves, providing effective control [44]. Isoprothiolane, a sulfur-containing systemic fungicide, shows strong control but should not be used alone long-term. Tebuconazole inhibits pathogen cell membrane formation, while Kasugamycin interferes with amino acid metabolism and mycelial growth [45]. Long-term and large-scale use of these single-component agents can lead to pathogen resistance and environmental concerns, including soil and water pollution and harm to non-target organisms [46]. Recent studies emphasize resistance monitoring using genomic tools to track *M. oryzae* mutations, guiding pesticide rotation strategies [47]. Integrated pest

management (IPM) frameworks, combining pesticides with cultural practices, have extended the efficacy of single-component agents [48].

Compound fungicides, such as Benzovindiflupyr + Azoxystrobin, Trifloxystrobin + Tebuconazole, and Propiconazole + Tricyclazole, act on multiple fungal biochemical pathways, thereby reducing the likelihood of resistance development in *M. oryzae* [49]. Multi-location field trials have reported control efficiencies of up to 90 % at optimal application rates, with synergistic interactions between active ingredients enhancing both efficacy and durability. Despite these benefits, concerns remain regarding environmental risks, including disruption of soil microbial communities, non-target toxicity, and potential water contamination [50]. To address these challenges, recent studies have focused on nano-formulations of compound fungicides [51], which improve active ingredient delivery, reduce application frequency, and minimize ecological impact while maintaining or even enhancing disease control effectiveness [52,53].

4.1.2. Biocontrol

Biocontrol generally refers to suppressing plant diseases using living organisms or their bioactive metabolites. In recent years, with the advancement of sustainable agriculture, stricter regulations on chemical pesticide use have driven the search for eco-friendly alternatives [54]. Compared with chemical fungicides, biocontrol agents and biopesticides are characterized by safety, environmental compatibility, and low toxicity, making them increasingly accepted in modern crop protection [55]. Biocontrol agents typically exert their effects through multiple mechanisms, including competition for nutrients and ecological niches, antibiosis, mycoparasitism, and the induction of host systemic resistance. This multi-target mode of action reduces the likelihood of resistance development in *M. oryzae* and minimizes chemical residues in agroecosystems [56]. For rice cultivation, the judicious use of biopesticides has become essential to ensure both sustainable production and stable yields [57,58]. Currently, rice biological pesticides can be broadly classified into plant-derived biopesticides, microbe-derived biopesticides and Phytoalexins, etc. A comprehensive list of biocontrol agents is provided in Supplementary Table S1.

Plant-derived compounds represent a rich source of natural antifungal agents against *M. oryzae*. For instance, neem (*Azadirachta indica*) seed extracts and saponins from *Schima superba* and *Sapindus mukorossi* have been shown to inhibit mycelial growth and spore germination of the pathogen [59]. Piperine from *Piper longum* demonstrated blast control efficacy comparable to 0.1 % Carbendazim, while leaving minimal environmental residues [60]. Recent advances in nano-encapsulation technologies have further improved

the stability, bioavailability, and field applicability of these phytochemicals, thereby enhancing their performance under variable environmental conditions [[61], [62], [63]].

Microbe-derived biopesticides represent one of the most extensively studied and promising biocontrol strategies for rice blast. Beneficial microorganisms such as *Bacillus subtilis*, *Pseudomonas fluorescens*, and *Trichoderma* spp. suppress pathogen development through competition, antibiosis, and mycoparasitism [64]. Field trials have shown that *P. fluorescens* can suppress blast incidence by up to 65 % in field trials through nutrient competition and induced systemic resistance [65], while *Trichoderma spp*. produce cell wall–degrading enzymes that reduce fungal pathogenicity by nearly 50 % [66]. Moreover, actinomycete-derived metabolites, identified via high-throughput screening, inhibit spore germination and appressorium formation, offering new biocontrol candidates [67]. Recent studies also highlight the potential of microbial consortia, in which synergistic interactions among multiple beneficial microbes further enhance biocontrol efficacy and consistency across environments [68].

In addition to exogenous agents, host-derived phytoalexins play an important role in biocontrol-related resistance enhancement. Key rice phytoalexins, including sakuranetin, oryzalexins A and B, and naringenin, exhibit potent inhibitory effects on *M. oryzae* infection [69]. Recent progress has clarified the biosynthetic pathways of these compounds, enabling targeted genetic engineering to increase phytoalexin accumulation in planta. Exogenous application of phytoalexin inducers, such as salicylic acid analogs and benzothiadiazole, has been reported to reduce rice blast incidence by up to 45 % under field conditions [70]. Moreover, CRISPR/Cas-based transcriptional activation of phytoalexin biosynthetic genes has shown promise in greenhouse trials, providing a novel route for durable resistance enhancement [71].

4.2. Cultivation of resistant varieties and molecular control strategies

4.2.1. Genetic resistance and breeding resources

Breeding rice varieties with durable resistance to *M. oryzae*, the causal agent of rice blast, remains a cornerstone of sustainable disease management. Although agronomic practices and chemical control can provide temporary relief, the high genetic variability and rapid evolution of *M. oryzae* highlight the urgent need for long-lasting genetic resistance in rice cultivars [72]. Significant progress has been made in elucidating the molecular basis of broad-spectrum and durable blast resistance. For example, recent studies have systematically identified and functionally characterized multiple resistance genes and immune signaling components, providing valuable genetic resources for resistance

breeding [73,74]. These advances reinforce the strategic importance of resistance breeding in achieving stable rice production and reducing reliance on chemical inputs.

To date, more than 140 rice blast resistance (*R*)-genes have been identified and mapped from both indica and japonica subspecies of rice, and about 36 genes have been molecularly characterized and cloned so far, which have boosted the breeding strategies for blast resistance in rice (Table 1) [74,75]. Among these cloned genes, several—namely *Pi9*, *Pi2*, and *Piz-t*—located on chromosome 6, are noted for conferring broad-spectrum resistance across multiple *M. oryzae* strains [76]. *Pi9*, in particular, confers the broadest spectrum of resistance and is thus of considerable breeding interest [77]. Furthermore, most studies focused on pyramiding individual *R* genes to develop single or multiple resistances, but such lines show less durability of resistance. Hence, a pyramiding of genes for a trait by combining two *R*-genes or more than two complementary *R*-genes could provide superior phenotypic benefits and stability compared to a single gene. However, the integration of molecular tools—such as marker-assisted selection (MAS), marker-assisted backcrossing (MABC), QTL mapping, and genome-wide association studies (GWAS)—has significantly accelerated the identification and pyramiding of resistance loci [75,78].

Table 1. Cloned blast resistance (*R*) genes/alleles in rice.

R_gene	**Encoding_protein**	**Chromosome**
Pi37	NLR	1
Pit	NLR	1
Pish	NLR	1
Pi35	NLR	1
Pi64	NLR	1
Pi-b	NLR	2
pi21	Proline-rich metal binding protein	4
Pi63/Pikahei-1(t)	NLR	4

R_gene	**Encoding_protein**	**Chromosome**
Pi9	NLR	6
Pi2	NLR	6
Piz-t	NLR	6
Pi-d2	B-lectin receptor kinase 6	6
Pi-d3	NLR	6
Pi25	NLR	6
Pid3-A4	NLR	6
Pi50	NLR	6
Pigm	NLR	6
Pi36	NLR	8
Pi5	NLR	9
Pii	NLR	9
Pi56	NLR	9
Pi54	NLR	11
Pikm	NLR	11
Pb1	NLR	11
Pik	NLR	11
Pik-p	NLR	11

R_gene	Encoding_protein	Chromosome
Pia	NLR	11
Pi1	NLR	11
Pi54rh	NLR	11
Pi-CO39	NLR	11
Pi54of	NLR	11
PiK-h	NLR	11
Pike	NLR	11
Piks	NLR	11
Pi-ta	NLR	12
Pb2	NLR	11

NLR, nucleotide-binding leucine-rich repeat. As cited in Wang et al. [74] and Sahu et al. [75].

Beyond classical *R* genes, regulatory resistance genes provide additional durability. The transcription factor allele *bsr-d1*, a novel allele of a C_2H_2-type transcription factor, confers variety-independent and persistent blast resistance by fine-tuning H_2O_2 homeostasis [79]. However, downstream components such as *Perox3* can negatively modulate this resistance when overexpressed, highlighting the complexity of redox-mediated immunity [80]. Moreover, it was found that *bsr-d1* also affects the biosynthesis and signaling network of salicylic acid - all of which are key components of systemic acquired resistance [81]. Another example is *RBR7* (Rice Blast Resistance 7), encoding a small 85-amino-acid protein; loss-of-function mutants show lesion mimic phenotypes, ROS accumulation, and elevated SA-responsive gene expression, leading to enhanced resistance to multiple blast isolates [82].

Importantly, resistance breeding has begun to reconcile the long-standing trade-off between yield and immunity. Another groundbreaking discovery involves IPA1 (Ideal

Plant Architecture 1, also known as SPL14) [83]. Typically recognized as a key regulator of rice architecture—controlling traits like tiller number, panicle branching, and lodging resistance—IPA1 was shown to harbor a gain-of-function allele (*ipa1-1D*) that not only boosts grain yield (>10 % higher than wild type) but also enhances resistance to rice blast [83]. Mechanistically, IPA1's switch between growth- and defense-promoting functions is mediated by its phosphorylation status: non-phosphorylated IPA1 activates downstream genes to promote yield, while phosphorylated IPA1 binds WRKY45 promoter motifs to trigger immune responses upon pathogen attack [84]. Remarkably, *ipa1-1D* maintains its yield advantage even under blast pressure, offering a rare allele that reconciles the typical trade-off between yield and disease resistance.

Together, these studies illustrate a multi-layered resistance framework that integrates *R* genes, regulatory alleles, and signaling modulators to achieve both agronomic performance and durable disease resistance.

4.2.2. Molecular control of rice blast and mechanistic insights

- (1)

Host Genome Editing Strategies (CRISPR/Cas9)

CRISPR/Cas9 technology has emerged as a powerful tool for engineering blast resistance by editing susceptibility genes, resistance genes, and regulatory elements. Multiplex editing of susceptibility genes such as *Pi21* and *ERF922* resulted in enhanced salicylic acid and jasmonic acid signaling and broad-spectrum resistance without severe yield penalties [85]. Similarly, targeted disruption of the microRNA gene *MIR827* abolished its expression and conferred blast resistance, revealing a trade-off between phosphate signaling and immunity [15].

Recent studies further demonstrate the potential of fine-tuned genome editing. An in-frame edited allele, $RBL1^{\Delta 12}$, generated through multiplex editing of the *RBL1* coding region, conferred strong resistance to multiple *M. oryzae* isolates with minimal developmental defects [86]. Building on this, AI-guided CRISPR/Cas9 promoter editing generated lines such as *Pro1*, which exhibit reduced *RBL1* expression, broad-spectrum resistance to rice blast, bacterial blight, and sheath blight, while maintaining wild-type yield [87]. Additionally, knockout of the phosphatidate phosphohydrolase (*PAH*) gene led to phosphatidic acid accumulation and enhanced blast resistance, linking lipid signaling to immune regulation [88]. These examples highlight that genome-level editing using CRISPR technology—by up-regulating resistance genes and down-regulating susceptibility genes—is an effective approach.

- (2)

RNAi-Based Approaches: HIGS and SIGS

RNA interference (RNAi) technologies offer complementary molecular strategies for blast control. Host-induced gene silencing (HIGS), in which rice plants express dsRNA targeting fungal genes such as *RGS1*, *MgAPT2*, and *LHS1*, significantly impairs fungal infection. It was shown that HIGS lines accumulated H_2O_2-rich defense granules and activated *PR* genes, resulting in robust resistance under greenhouse and field conditions [16]. Earlier and complementary studies by Wang and Dean (2022) confirmed that HIGS targeting development and pathogenicity genes in *M. oryzae* suppresses fungal growth and can be used as a functional approach to validate virulence factors [89].

Spray-induced gene silencing (SIGS) offers a non-transgenic alternative, applicative route by foliar-applying dsRNA or dsRNA-formulations that are taken up by the fungus or plant and trigger sequence-specific silencing. Evidence from Sarkar and Roy-Barman (2021) has shown direct foliar sprays of naked dsRNA can silence *M. oryzae* pathogenicity genes (e.g., *MoDES1*) and reduce infection, demonstrating in planta uptake and systemic transfer of silencing signals in the rice–blast pathosystem [90]. The recent report by Ladera Carmona et al. (2025) showed that foliar application of dsRNA targeting *M. oryzae* genes effectively suppressed fungal growth through RNAi-mediated degradation and stress responses [91]. However, naked dsRNA is rapidly degraded in the environment; thus, recent advances emphasize carrier systems — notably nanoparticles and polymer encapsulation — to protect dsRNA, improve uptake, and prolong activity [92]. Polymer-encapsulated dsRNA delivered via foliar application provided prolonged gene knockdown and durable protection in trials, and several reviews and experimental studies confirm that nano-formulations markedly increase SIGS efficacy against fungal pathogens [93].

Together, HIGS provides durable, target-specific resistance and a powerful research tool to validate fungal genes, but it requires transgenic lines and faces regulatory and public-acceptance barriers. SIGS is transgene-free and adaptable to integrated pest management, yet its success hinges on stable, cost-effective delivery systems; nanocarrier technologies (polymeric encapsulation, clay/liposome matrices) might be the most promising path to field-scale application. Recent global work therefore converges on hybrid strategies (e.g., HIGS for breeding and SIGS for emergency/seasonal protection), coupled with genomic surveillance to select optimal target genes and minimize resistance evolution.

- (3)

Effector Biology and Host Immunity

Deciphering effector–host interactions has become central to understanding rice blast resistance and developing durable control strategies. *M. oryzae* deploys a diverse repertoire of secreted effector proteins to suppress host immunity, manipulate cellular processes, and facilitate colonization. In response, rice has evolved multilayered immune systems, including effector-triggered immunity (ETI) mediated by NLR proteins and quantitative resistance mechanisms that fine-tune basal defense responses [94]. Recent molecular dissection of resistance pathways revealed downstream signaling modules such as the PiBP1–WAK14–PAL1 relay, which amplifies defense responses upon pathogen recognition [16]. In parallel, quantitative resistance loci (*qR* genes), including *Pi21* and *Bsr-d1*, confer partial but durable resistance by regulating ROS accumulation and reinforcing cell wall integrity [85]. Another example is the identification of a fungal secreted protein (MoSPAB1) that directly activates the rice transcription factor BSR-D1, promoting peroxidase expression and H_2O_2 scavenging to favor infection — a clear case of an effector manipulating host regulatory circuitry to induce susceptibility. This study links effector action to a specific host susceptibility node and explains how natural variants in the *bsr-d1* promoter confer durable resistance [95]. Importantly, effectors that hijack qR-linked regulators (as with MoSPAB1–BSR-D1) reveal mechanistic crosstalk between ETI and quantitative resistance, supporting strategies that combine *R*-gene stacking with qR exploitation [96]. These studies underscore the importance of integrating major *R* genes with quantitative resistance mechanisms to enhance durability.

- (4)

Omics and Functional Genomics–Guided Resistance Discovery

Emerging omics technologies are fundamentally reshaping the discovery and deployment of disease resistance in rice. Advances in genomics and synthetic biology have enabled the rational engineering of synthetic NLR immune receptors with broadened and customizable effector-recognition spectra, providing a proactive strategy to counter the rapid evolution of *M. oryzae* effectors [97]. In parallel, rice pan-genome analyses have revealed extensive presence–absence variation of resistance genes, highlighting the limitations of single-reference genomes and uncovering novel NLRs and resistance-associated loci absent from elite cultivars [98]. Transcriptomic approaches have also evolved from bulk RNA sequencing to single-cell resolution. Single-cell RNA sequencing (scRNA-seq) of rice leaves during early *M. oryzae* infection has uncovered strong cell-type–specific immune responses, including distinct defense programs in

epidermal, mesophyll, and vascular-associated cells, and has led to the identification of previously unrecognized immune regulators [99]. When integrated with spatial transcriptomics, these approaches provide unprecedented insights into the spatial organization of host–pathogen interactions.

Notably, functional genomics tools have accelerated the translation of omics-derived candidates into validated resistance genes. CRISPR/Cas-based perturbation screens, including multiplex and pooled editing strategies, enable high-throughput functional interrogation of candidate genes involved in immune signaling and susceptibility [100]. Genome-wide association studies (GWAS), combined with transcriptome-wide association studies (TWAS), have identified numerous QTLs and SNPs linked to both yield and disease resistance, facilitating the pyramiding of resistance and agronomic traits [101]. Anantharasa et al. (2025) further emphasized that systematic assessments of morphotype and genetic diversity are essential for identifying elite resistance donors from diverse germplasm pools [102].

Together, resistance gene discovery, genome editing, RNAi technologies, effector biology, and omics-driven functional genomics form an integrated molecular toolbox for rice blast management. Rather than acting independently, these approaches increasingly converge—CRISPR enables precise manipulation of *R* and *qR* genes, RNAi provides flexible pathogen-targeting strategies, and omics technologies guide rational design. This integrated framework promises to enhance resistance durability, reduce fungicide dependence, and support sustainable rice production under evolving pathogen and climate pressures.

5. Summary and outlook

M. oryzae continues to pose a major threat to global rice production, with profound impacts on food security and economic stability. Current control strategies primarily rely on chemical fungicides and resistant varieties, face increasing limitations due to environmental concerns and the rapid evolutionary adaptability of the pathogen. These challenges highlight the urgent need for more durable and sustainable control approaches. Significant research gaps persist, including a deeper understanding of pathogen population dynamics, improving the scalability of biological control agents, and developing resistance that remains effective under diverse environmental conditions. Recent advances in molecular breeding, particularly gene pyramiding and CRISPR/Cas9-mediated genome editing, provide promising opportunities to enhance broad-spectrum and long-lasting resistance. In parallel, improved pathogen surveillance using high-

throughput sequencing and qPCR technologies will be essential for tracking virulence evolution and guiding timely interventions [103].

Looking forward, a transformative direction lies in the deeper integration of field-scale big data with advanced AI-driven intelligent models. These tools can facilitate real-time monitoring of rice blast incidence across diverse regions, rapid assessment of resistance gene efficacy in deployed cultivars, and predictive forecasting under climate variability. Ultimately, a prevention-oriented, integrated control framework—combining durable resistant varieties, optimized agronomic practices, precise and targeted fungicide application, and cultural measures like crop rotation and silicon fertilization—offers the most practical and effective path forward. By proactively addressing pathogen evolution and climate change impacts within this holistic, data-informed strategy, future rice blast management will be better equipped to sustain rice productivity and safeguard global food security.

CRediT authorship contribution statement

Fei Xu: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Zheng-Quan He:** Resources, Investigation, Formal analysis. **Ming-Xing Cheng:** Writing – review & editing. **Fei-Fei Li:** Writing – review & editing. **Lu-Lu Yu:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare no conflicts of interest.

Appendix A. Supplementary data

The following is the Supplementary data to this article: